



UMT

Computer Network Project :

Web Traffic Monitoring and Network Security

Report:

PARTICIPANT NAME	PARTICIPANT ID'S
ALE E EBA	F2024105020
AMNA ARSHAD	F2024105022
AYMA WAHEED	F2024105048
SUBMITTED TO	
Mam.Shanza Zaman	
SESSION	
Y3	

ABSTRACT

This project demonstrates a secure network monitoring and management system using Cisco Packet Tracer and Wireshark. The project focuses on monitoring HTTP web traffic and implementing network security techniques such as SSH (Secure Shell), Access Control Lists (ACLs), password encryption, and switch port security.

A hotel booking website is hosted on a web server where users submit booking information through an HTTP webpage. The project demonstrates how HTTP traffic can be monitored and captured using Packet Tracer Simulation Mode and Wireshark, showing that sensitive information is transmitted in plaintext. Security measures are then applied to protect the network infrastructure and demonstrate the importance of secure communication methods such as SSH and HTTPS.

INTRODUCTION

Computer networks are essential for communication and data sharing in modern organizations. However, network security is necessary to protect devices, user information, and communication from unauthorized access and cyber threats.

This project demonstrates a hotel network environment where users access a booking website hosted on a server. Since HTTP traffic is not encrypted, sensitive information entered by users can be captured and viewed in plaintext. To improve network security, SSH is configured on the Cisco router for secure remote administration, ACLs are implemented to restrict unauthorized access, and switch port security is enabled to protect the network from unauthorized devices.

The project highlights the importance of using secure protocols such as SSH instead of Telnet and HTTPS instead of HTTP.

PROJECT OBJECTIVES

The main objectives of this project are:

- To create a secure network environment using Cisco Packet Tracer
- To monitor HTTP traffic and observe unencrypted data transmission
- To configure SSH for secure remote router management
- To implement password encryption for router security
- To apply Access Control Lists (ACLs) for restricted access
- To configure switch port security to prevent unauthorized access

- To demonstrate the difference between HTTP and HTTPS
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PROJECT GOALS

This project demonstrates how to improve network security using:

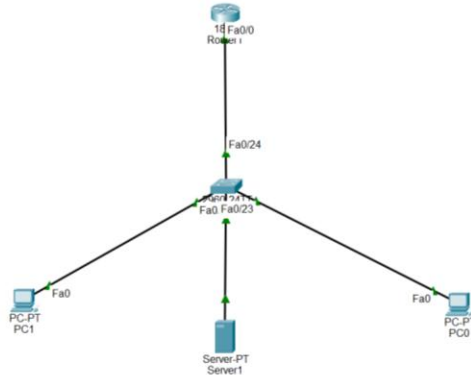
- **SSH** for secure remote router management
 - **Router password encryption** to secure credentials
 - **Access Control Lists (ACLs)** to restrict unauthorized access
 - **Port Security** on switches to prevent MAC spoofing and unauthorized devices
 - **DNS Services** for domain name access
 - **HTTP/HTTPS Services** for website accessibility and secure communication
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NETWORK TOPOLOGY

The network topology consists of the following devices:

- **1 Router – Cisco 1841**
Used for routing and secure remote administration through SSH.
- **1 Switch – Cisco 2960-24TT**
Connects all devices and implements switch port security.
- **1 Web Server – Server-PT**
Hosts the hotel booking website using HTTP and HTTPS services.
- **2 PCs – PC0 and PC1**
Used by users to access the website and test network connectivity.

All devices are connected using **copper straight-through cables**.



DEVICES USED AND IP ADDRESSING

The following IP addressing scheme was configured in the project:

Device IP Address

Router 10.10.10.1

Server 10.10.10.10

PC0 10.10.10.20

PC1 10.10.10.21

Subnet Mask: 255.0.0.0

Default Gateway: 10.10.10.1

DNS Server: 10.10.10.10

HTTP SERVICE CONFIGURATION

The web server was configured to provide HTTP and HTTPS services.

The following steps were performed:

1. Open the server in Cisco Packet Tracer.
2. Navigate to **Services** → **HTTP**.
3. Turn **HTTP and HTTPS ON**.
4. Create the following files:
 - **index.html** for the main booking webpage
 - **book.php** for booking confirmation after form submission

The hotel booking website allows users to enter booking information through a form.

The image shows two side-by-side browser windows. The left window displays the 'Hotel Paradise' booking form. It has a title 'Hotel Paradise' and a subtitle 'Secure Room Booking'. The form includes fields for a search bar (containing 'all'), an email field (containing 'lal@gmail.com'), a room type dropdown (selected 'Suite Room - \$200/night'), check-in and check-out dates (both '06/18/2024'), a phone number field (containing '1122-9900-3322-8899'), a password field (containing '****'), and a CVV field (containing '*****'). A 'Book Now' button is at the bottom. A red warning message at the bottom of the form states 'HTTP MODE: Your data is NOT encrypted!'. The right window shows the 'Booking Confirmed!' page. It features a green checkmark icon, the text 'Booking Confirmed!', and a thank you message 'Thank you for choosing Hotel Paradise'. Below this is a 'Booking Details' section with fields for Name, Email, Room Type, Check-in, Check-out, Credit Card, and CVV. A red warning message at the bottom of the details section states 'SECURITY WARNING: This data was sent via HTTP! Your credit card information is visible in Wireshark!'. A 'Back to Home' button is at the bottom of the page.

DNS SERVICE CONFIGURATION

DNS service was configured to allow users to access the website through a domain name.

Configuration Steps:

1. Open **Server** → **Services** → **DNS**
2. Enable DNS service
3. Add the domain name:

hotel.com → **10.10.10.10**

This allows users to access the website through:

- **IP Address:** 10.10.10.10
- **Domain Name:** hotel.com

HTTP ACCESS TESTING

The website was successfully accessed through:

IP Access

Users accessed the website using:

<http://10.10.10.10>

Domain Access

Users also accessed the website through:

<http://hotel.com>

This verified that DNS services were functioning correctly.

SIMULATION MODE BEFORE SSH

Before implementing network security, HTTP traffic was monitored using **Packet Tracer Simulation Mode** and **Wireshark**.

Procedure:

1. Open Simulation Mode or Wireshark
2. Apply the **HTTP filter**
3. Submit the booking form
4. Capture the transmitted packets

The captured HTTP packets showed that user information was visible in plaintext.

This demonstrated that HTTP communication is **not encrypted**, making sensitive information vulnerable to attackers.

ACCESS CONTROL LIST (ACL) CONFIGURATION

An Access Control List (ACL) is a network security feature used to control and filter network traffic. ACLs help administrators permit or deny access to specific devices, networks, or services based on security requirements.

In this project, an ACL was configured on the Cisco router to restrict unauthorized access and improve network security. The ACL ensured that only authorized devices from the permitted subnet could access network resources, while unauthorized traffic was blocked.

The ACL was implemented to:

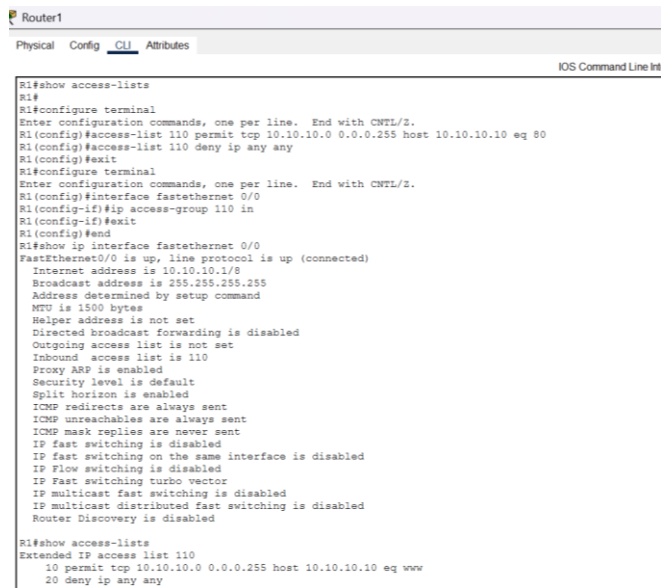
- Restrict unauthorized access to the web server
- Improve network security

- Control traffic flow between devices
- Allow only permitted users to access network resources

Benefits of ACL

- Improves network security
- Prevents unauthorized access
- Filters unwanted traffic
- Controls communication between devices
- Protects network resources

The ACL configuration successfully helped secure the network by controlling which devices were allowed to communicate within the network.



```

Router1
Physical Config CLI Attributes
IOS Command Line Interface

R1#show access-lists
R1#
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#access-list 110 permit tcp 10.10.10.0 0.0.0.255 host 10.10.10.10 eq 80
R1(config)#access-list 110 deny ip any any
R1(config)#exit
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface fastethernet 0/0
R1(config-if)#ip access-group 110 in
R1(config-if)#exit
R1(config)#end
R1#show ip interface fastethernet 0/0
FastEthernet0/0 is up, line protocol is up (connected)
Internet address is 10.10.10.1/8
Broadcast address is 255.255.255.255
Address determined by setup command
MTU is 1500 bytes
Helper address is not set
Directed broadcast forwarding is disabled
Outgoing access list is not set
Inbound access list is 110
Proxy ARP is enabled
Security level is default
Split horizon is enabled
ICMP redirects are always sent
ICMP unreachable are always sent
ICMP mask replies are never sent
IP fast switching is disabled
IP fast switching on the same interface is disabled
IP flow switching is disabled
IP Fast switching turbo vector
IP multicast fast switching is disabled
IP multicast distributed fast switching is disabled
Router Discovery is disabled

R1#show access-lists
Extended IP access list 110
10 permit tcp 10.10.10.0 0.0.0.255 host 10.10.10.10 eq www
20 deny ip any any
  
```

ROUTER CONFIGURATION

The Cisco router was configured to improve network security.

The following security configurations were applied:

- **Hostname Configuration (R1)**
- **Password Encryption**
- **SSH Version 2 Configuration**
- **Local User Authentication**
- **Domain Name Configuration**

- **ACL Implementation**

SSH was configured to allow secure remote access to the router and prevent unauthorized management access.

```
Router1
Physical Config CLI Attributes
IOS Command Line Interface
R1(dhcp-config)#default-router 10.10.10.1
R1(dhcp-config)#dns-server 8.8.8.8
R1(dhcp-config)#exit
R1(config)#! SSH configuration
R1(config)# ip domain-name hotel.local
R1(config)#crypto key generate rsa
% You already have RSA keys defined named R1.hotel.local :
% Do you really want to replace them? (yes/no): yes
The name for the keys will be: R1.hotel.local
Choose the size of the key modulus in the range of 360 to 2048 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes:
How many bits in the modulus (512): 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]
R1(config)#ip ssh version 2
*Mar 3 0:29:52.181: %SSH-5-ENABLED: SSH 2 has been enabled
R1(config)#line vty 0 4
R1(config-line)#username admin secret admin123
R1(config-line)#login local
R1(config-line)#
% Invalid input detected at '^' marker.
R1(config)#line vty 0 4
R1(config-line)#
% Invalid input detected at '^' marker.
R1(config)#line vty 0 4
R1(config-line)#transport input ssh
R1(config-line)#login local
R1(config-line)#exit
R1(config)#service password-encryption
R1(config)#end
R1#
sys-5-CONFIG-1: Configured from console by console
copy running-config startup-config
Destination filename: [startup-config]?
Building configuration...
[OK]
R1#
```

SWITCH CONFIGURATION

Switch security was configured to protect the network from unauthorized devices.

Security Features Implemented:

- Port Security
- MAC Address Restriction
- Prevention of Unauthorized Device Access

This helps secure the local network and reduces security risks.


```
Switch0
Physical Config CLI Attributes
IOS Command Line Interface
floor1(config-if)#switchport mode access exit
% Invalid input detected at '^' marker.
floor1(config-if)#switchport mode access
floor1(config-if)#exit
floor1(config)#end
floor1#
%SYS-5-CONFIG_I: Configured from console by console
copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
floor1#show port-security
Secure Port MaxSecureAddr CurrentAddr SecurityViolation Security Action
-----
Fa0/1 1 1 1 Restrict
Fa0/2 1 0 0 Restrict
Fa0/3 1 1 0 Shutdown
floor1#show port-security access
% Invalid input detected at '^' marker.
floor1#show port-security access
% Invalid input detected at '^' marker.
floor1#show port-security address
Secure Mac Address Table
-----
Vlan Mac Address Type Ports Remaining Age
----
10 0005.SEC0.DA01 SecureSticky Fa0/1 -
1 0060.SEC0.9B94 DynamicConfigured FastEthernet0/3 -
Total Addresses in System (excluding one mac per port) : 0
Max Addresses limit in System (excluding one mac per port) : 1024
floor1#
```

PING TESTING

Connectivity tests were performed between devices using the **ping command**.

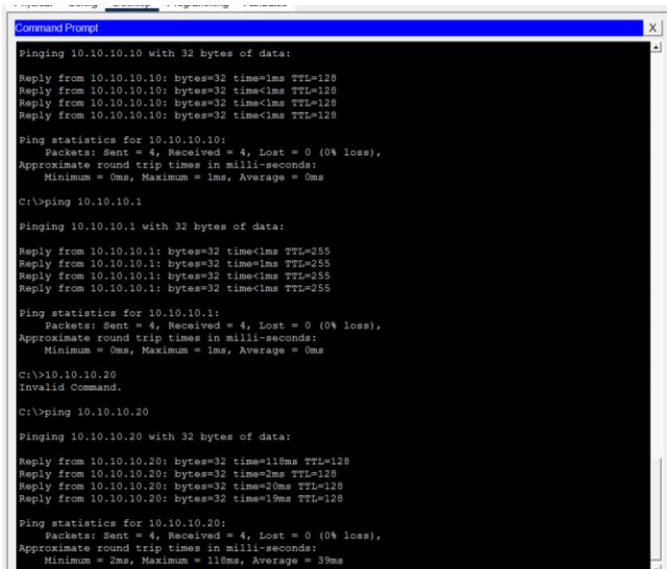
Ping from PC1:

- Server (10.10.10.10) — Successful
- Router (10.10.10.1) — Successful
- PC0 (10.10.10.20) — Successful

Ping from PC0:

- Server (10.10.10.10) — Successful
- Router (10.10.10.1) — Successful
- PC1 (10.10.10.30) — Successful

The successful ping results confirmed proper network connectivity.



```
Command Prompt
Pinging 10.10.10.10 with 32 bytes of data:
Reply from 10.10.10.10: bytes=32 time<1ms TTL=128
Reply from 10.10.10.10: bytes=32 time<1ms TTL=128
Reply from 10.10.10.10: bytes=32 time<1ms TTL=128
Reply from 10.10.10.10: bytes=32 time<1ms TTL=128
Ping statistics for 10.10.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 10.10.10.1
Pinging 10.10.10.1 with 32 bytes of data:
Reply from 10.10.10.1: bytes=32 time<1ms TTL=255
Reply from 10.10.10.1: bytes=32 time<1ms TTL=255
Reply from 10.10.10.1: bytes=32 time<1ms TTL=255
Reply from 10.10.10.1: bytes=32 time<1ms TTL=255
Ping statistics for 10.10.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 10.10.10.20
Invalid Command.
C:\>ping 10.10.10.20
Pinging 10.10.10.20 with 32 bytes of data:
Reply from 10.10.10.20: bytes=32 time=118ms TTL=128
Reply from 10.10.10.20: bytes=32 time=2ms TTL=128
Reply from 10.10.10.20: bytes=32 time=28ms TTL=128
Reply from 10.10.10.20: bytes=32 time=19ms TTL=128
Ping statistics for 10.10.10.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 118ms, Average = 39ms
```

SIMULATION MODE AFTER SSH

After implementing SSH, router management traffic became encrypted.

Procedure:

1. Open Simulation Mode
2. Apply filters
3. Capture packets
4. Observe network traffic

SSH protected remote administration by encrypting router login communication, preventing attackers from viewing login credentials.

When HTTPS was enabled, web traffic also became encrypted, protecting user information from plaintext exposure.

ADVANTAGES OF SSH OVER TELNET

SSH (Secure Shell) is more secure than Telnet because it encrypts communication between devices.

Advantages of SSH:

- Provides encrypted communication
- Protects usernames and passwords

- Prevents unauthorized access
- Supports secure remote administration
- More secure than Telnet

Telnet sends data in plaintext, making it vulnerable to attacks and credential theft.

DIFFERENCE BETWEEN HTTP AND HTTPS

HTTP	HTTPS
Data is unencrypted	Data is encrypted
Less secure	More secure
Uses Port 80	Uses Port 443
Vulnerable to attacks	Protects user data

HTTP does not encrypt transmitted data, while HTTPS provides secure encrypted communication between users and servers.

CONCLUSION

This project successfully demonstrated a secure network monitoring and management system using Cisco Packet Tracer. Security techniques such as SSH remote access, password encryption, ACLs, switch port security, HTTP/HTTPS services, and DNS configuration were implemented to improve network protection.

The project showed how HTTP traffic can expose sensitive information in plaintext and highlighted the importance of secure communication protocols. It also emphasized the use of SSH instead of Telnet and HTTPS instead of HTTP to improve security in modern computer networks.

Overall, the project provided practical knowledge of network security, monitoring, and secure remote administration.